A Study on Risk and Return Analysis of Axis Bank Ltd Listed on the National Stock Exchange: An Empirical Study Using CAPM Model (2021-2024)

Submitted in partial fulfilment of

"Post-Graduation Diploma in Management"

Under the Esteemed Guidance of

Dr. K. Subramani

Submitted by

Mittapalli Lakshmi Naga Kavya

Roll No:241357

Batch: 2024-26

Vignana Jyothi Institute of Management

(Approved by AICTE, Ministry of HRD, (Govt. of India)

Bachupally, Hyderabad

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PGDM BATCH 2024-26

Submitted to
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Student Declaration

I, Mittapalli Lakshmi Naga Kavya, (Roll No. 241357) hereby declare that the project titled "A Study on Risk and Return Analysis of Axis Bank Ltd Listed on the National Stock Exchange: An Empirical Study Using CAPM Model (2021-2024)" is an original work undertaken by me, under the guidance of Dr. K. Subramani.

The report submitted is a Bonafide work of my own efforts and has not been submitted to any institute/university/conference or published before.

Signature of the student

Mittapalli Lakshmi Naga Kavya

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Faculty Guide Certificate

I, Dr. K. Subramani. certify that Ms. Mittapalli Lakshmi Naga Kavya from Batch 2024-26 bearing Roll No.241357 has done the work under my guidance and the report presented is genuine to the best of my knowledge.

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Introduction

Overview of Indian Banking Sector

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I, Mittapalli Lakshmi Naga Kavya, the student, would like to express our deepest gratitude to everyone who made our dissertation project a truly enriching and rewarding experience.

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I would also like to thank my faculty at **Vignana Jyothi Institute of Management** for preparing me to take on this challenge with confidence. Your academic foundation empowered me to translate theoretical knowledge into practical skills during this dissertation.

This experience has not only strengthened my research skills but also broadened my perspectives, and I am truly thankful for every interaction, lesson, and opportunity I encountered along the way.

I look forward to carrying these learnings into my future endeavours and hope to stay connected with everyone who made this journey so memorable.

Thank you once again!

Name of the Student: Mittapalli Lakshmi Naga Kavya



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Executive Summary

As part of the PGDM curriculum at Vignana Jyothi Institute of Management, Bachupally, Hyderabad, I successfully completed my dissertation project. This dissertation provided me with a practical platform to apply my academic learning in a real-world business environment and gain valuable industry exposure.

The primary objective of our project was to [briefly state the main project goal — e.g., "Challenges and Prospects of fintech or "optimize the supply chain process to enhance operational efficiency"]. Over the course of the dissertation, I conducted extensive research, engaged in data analysis, collaborated with cross-functional teams, and developed actionable recommendations.

Key outcomes of the project include:

- **First major result or finding**

  The study looked at the relationship between risk and return of Axis Bank Ltd stock listed on the National Stock Exchange of India. It used the Capital Asset Pricing Model (CAPM) for the period from 2021 to 2024. By analyzing historical stock price data and market index returns, the research helped us understand how market fluctuations impact the performance of Axis Bank shares and how risk and return relate within the banking sector.

- **Second major result or finding**

  The research estimated the systematic risk (Beta) of Axis Bank stock using statistical tools and regression analysis. The calculated beta value indicated that Axis Bank is more sensitive to market movements compared to the overall market index. This finding shows that the stock has higher market risk and tends to respond more strongly to changes in economic and market conditions.

- **Third major result or contribution**

  The study compared the expected return calculated with the CAPM model to the actual average return of Axis Bank during the study period. The results revealed that the actual return was slightly lower than the expected return predicted by CAPM. This indicates that the stock marginally underperformed relative to its level of risk. This outcome provides helpful insights for investors evaluating the risk and return of banking sector stocks.

Throughout the dissertation, I enhanced my skills in particularly research skills Financial analysis skills, Data analysis using Excel, Research and analytical skills, Understanding of CAPM and risk-return relationship, Interpretation of financial data, Report writing and academic research skills and gained deeper insights into Banking Sector.

I am grateful for the guidance and mentorship provided by Dr. K. Subramani, who greatly contributed to the successful completion of my project. I also thank **Vignana Jyothi Institute of Management**, for equipping me with the necessary knowledge and skills to take on this challenge.

This dissertation project experience has been helpful in bridging the gap between theoretical concepts and industry practices, and it has significantly shaped my professional development as future business leaders.

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CHAPTER 1: INTRODUCTION

Introduction

The idea of risk and return is a basic principle of financial management and investment decisions. Every investment is associated with a level of uncertainty in terms of future returns, price fluctuations, and market conditions. On a broader level, investors are ready to take a high level of risk if there is a possibility of earning high returns. Hence, it is a basic principle for investors to understand the relationship between risk and return in order to make rational and informed decisions in the capital market.

In recent times, the Indian stock market has witnessed tremendous growth in terms of market efficiency. The use of digital platforms for trading, a rise in retail investors, a high level of financial literacy among investors, and proper supervision by market regulators have all contributed to the efficiency of the market.

Among all the companies listed on the NSE, Axis Bank Ltd is one of the top private sector banks that is also part of the Nifty 50 list. Due to its presence in the financial sector of the Indian economy, significant to understand risk and return of profile of Axis Bank Ltd from an investor's perspective who is interested in investing in banking sector stocks.

The objective of this study is to analyze the risk-return profile of Axis Bank Ltd with the help of various financial and statistical tools such as Standard Deviation, Beta, and CAPM. It is important to understand whether the returns earned from investing in the stock of Axis Bank Ltd are justified or not on the basis of the level of risk taken. It is expected that the findings of this study would be able to provide investors with an idea of the investment potential of Axis Bank Ltd in the current market environment.

Overview of Indian Banking Sector

The banking sector is of critical importance to the economic development of India. The banks are responsible for mobilizing savings from the public and directing them towards investments in productive sectors in the form of credits to individuals, businesses, and industries. This way, banks are significantly contributing to the development of India's economy. Out of these categories of banks, the private sector banks have emerged as the major competitors compared to the others in terms of efficient banking, the use of technology, and the quality of customer service.

These include public sector banks, private sector banks, foreign banks, and small finance banks. Among these, private sector banks are growing as strong competitors. Private sector banks are

emphasizing efficient operations, technology adoption, and customer service. Hence, they are growing faster in retail lending, technology adoption, and profitability compared to traditional banks.

Any fluctuation in repo rates, liquidity, and regulations would directly impact profitability and asset quality of banks and hence impact banking stocks. So, banking stocks are also impacted by overall market risks.

The economic recovery from the pandemic, digital transactions, and financial sector reforms have further altered the banking scenario. Investors are highly interested in banking stocks since they are a reflection of the overall stability and growth of the economy. Hence, by studying the risk-return characteristics of a leading private sector bank, we can gain a deeper insight into the performance of the banking sector and investment opportunities in the Indian capital market.

Overview of Axis Bank LTD

One of the top banks in India's private banking sector, Axis Bank Ltd. is included in the Nifty 50 stock index. Since its founding in 1993, it has increased its footprint in international business, corporate banking, treasury operations, and retail banking. Axis Bank has improved its standing in the Indian stock market throughout time through a number of financial products, acquisitions, and technological developments.

Axis Bank serves a wide range of customers, including small and medium-sized businesses, major corporations, and retailers, with banking and financial services. It has maintained its competitiveness in India's private sector banking market by putting more of an emphasis on digital banking and data-based financial solutions.

Axis Bank's stock performance is affected and reflects the market trends because it is included in the Nifty 50 list. Axis Bank's stock is also held by many normal investors, institutional investors, and mutual investors, and hence is an excellent stock for carrying out research on risk and return.

Axis Bank Ltd is a suitable company for carrying out research on the risk-return relationship because of its high market value and liquidity and its importance in the financial system. Statistical tools such as Standard Deviation, Beta, and CAPM can be employed for carrying out research on the risk-return relationship and checking whether the returns are commensurate with the risks taken.

Background of the Study

Investment risk is of two kinds: systematic risk and unsystematic risk. Systematic risk is also known as market risk. It arises from various economic factors such as fluctuations in interest rates, inflation, economic recession, and other market fluctuations. It is a kind of risk that affects all securities available in the market. It is not possible to avoid this kind of risk through diversification. On the other hand, unsystematic risk is related to a specific company and may result from decisions made by its management or problems related to a specific industry.

Unlike systematic risk, unsystematic risk can be managed with a diversified portfolio. Only systematic risk is considered to be an important factor for investment return by the Capital Asset Pricing Model. Hence, this study is focused on market-related risk, also known as Beta.

Risk and return are two important topics in investment and financial. Risk is the possibility that the actual return on an investment may vary from the expected return. Such uncertainty is due to various factors that are linked to the economy, markets, and businesses. In financial markets, risk generally refers to fluctuations in investment returns.

Risk is of two types: systematic and unsystematic. Systematic risk, also called market risk, is linked to various factors like inflation, fluctuations in interest rates, political problems, and market trends. Such risks affect the whole market and cannot be managed in any way. On the other hand, unsystematic risks are those that are unique to a particular company or industry.

Return can also be in the form of capital appreciation, in which case the value of a stock increases. Return can also take the form of dividends paid to shareholders of a company. Investors seek to analyze past returns to establish patterns and determine whether or not they are getting enough returns in proportion to the risks involved in an investment. To scientifically analyze this, models such as the CAPM are used.

CAPM formula:

$$\textbf{Expected Return} = R_f + \beta\,(R_m - R_f)$$

Where

R_f represents the risk-free rate

β represents systematic risk

$(R_m - R_f)$ represents the market risk premium.

Research Problem

Investors willing to invest their funds in banking sector, considering growth prospects, stability of dividends, and their contribution to economic growth. This sector attracts both individual and institutional investors due to its size, market value, and its ability to pay dividends.

The stock of the companies in the banking sector can be very volatile and sensitive to external factors they are changes in interest rates, inflation, credit growth, and other market conditions. This makes it difficult to predict the returns, thereby creating confusion for investors while making decisions regarding their investments.

It is difficult to ascertain whether the returns obtained from the particular stock of the banking company are enough to cover the risk taken by the investors without conducting any scientific or quantitative analysis of the relationship between systematic risk and expected returns.

Thus, the basic question that is being answered in this research is: Does Axis Bank Ltd generate returns in line with its level of systematic risk as predicted by the CAPM?

Research Questions

1. What is the average return earned by Axis Bank Ltd during the specified period of time?

2. What is the total risk as per standard deviation?

3. What is the beta value of Axis Bank relative to the market index?

4. What is E(R) of Axis Bank as per CAPM?

5. Is Axis Bank fairly valued based on actual and expected returns?

Objectives of the Study

1. To calculate historical returns earned by Axis Bank Ltd.
2. To calculate total volatility as per standard deviation.
3. To calculate systematic risk.
4. To calculate E(R) using CAPM.
5. To compare actual and expected returns using CAPM.

Relevance of the Study

The importance of the present study arises from the ups and downs and changes in the Indian capital market over the years. The economy has been recovering from the pandemic, the Reserve Bank of India (RBI) is changing the interest rate, and inflation and economic

uncertainties are affecting the stock market. Under such circumstances, the relationship between risks and returns has to be understood to make smart investments.

The banking industry is significant in the context of the overall financial market. The banking industry's stocks are more volatile and respond to changes in the interest rate, credit growth, liquidity, and government policies. Axis Bank Ltd, being one of the major private sector banks, which forms part of the Nifty 50 index, represents the financial industry and the market as a whole.

Besides, more retail investors are entering the Indian stock exchange with the help of digital trading platforms and through financial literacy. Nevertheless, investors rely more on overall stock exchange trends instead of using financial theories to evaluate stock performance. This research aims at demonstrating how investors can use financial theories like the capital asset pricing model in making investment decisions. This research is relevant in today's stock exchange market since it provides investors with a new perspective on the risk and return relationship of Axis Bank Ltd.

Scope of the Study

The research has utilized secondary data from reliable sources like the National Stock Exchange (NSE). The research aims to analyze historical data regarding the stock price of Axis Bank Ltd over a period of four years. The monthly data will be helpful in calculating average returns, standard deviations, and beta values. The research aims to evaluate the systematic risk and average returns using the CAPM. The research has not considered multi-factor asset pricing models, which are Arbitrage Pricing Theory (APT) and Fama-French model. The research has only considered market-related risk; it has not considered company-related risks. The findings are based on the data considered over a specific time period and are valid under changing economic conditions and time frames.

Limitation of the Study

1. Limited Period

In this study which is four years, ranging from January 2021 to December 2024. The market environment may change over a longer period of time. Hence, the results may not be representative of the long-term risk and return properties.

2. Dependence on Secondary Data

The analysis is based solely on secondary data obtained from financial websites, such as stock market websites. Any errors or inaccuracies in the data may affect the results obtained.

3. Use of a Single Company

The study is based on Axis Bank Ltd, which is listed on the National Stock Exchange. Hence, it is not applicable to all banks and companies in the Indian stock exchange.

4. Fundamentals of CAPM Model

The CAPM model is based on various assumptions. For example, it assumes that markets are efficient and that investors behave in a rational way. Also, it assumes perfect knowledge of markets. These may not be actual conditions.

5. Market Proxy Limitation

The research utilized the Nifty 50 index as a proxy to measure the general market's return. It is possible that this proxy does not entirely capture the stock market's performance.

6. Assumption of Constant Risk-Free Rate

The research assumed a constant risk-free rate based on the average RBI 91-day Treasury Bill rate. In reality, this is not constant and varies over time, which may impact the results of CAPM.

CHAPTER 2: LITERATURE REVIEW

2.1 Review of Existing Theories and Models

The study is based on well-known theories of finance that illustrate how risk influences return in financial markets. It is essential to comprehend these theories to evaluate the performance of Axis Bank Ltd CAPM. The next section of this study is a detailed discussion of major theories related to this research.

Concept of Risk and Return

Two factors are involved in taking the right decision regarding investments: risk and return. The return is the profit or loss on an investment over a particular period of time, expressed as a percentage. Risk is the uncertainty about achieving the expected return.

In the financial market, most investors try to avoid taking risks. Investors are only willing to take more risks if they are offered higher returns to compensate for the risk.

In the case of banking stocks such as Axis Bank Ltd., risks can arise from different factors, such as:

Credit risk

Regulatory policy

Economic slowdown

Understanding the relationship between risk and return is important to determine whether Axis Bank stock is providing the right return for the risk taken.

Types of Risk in Financial Markets

There are two types of risks in the financial markets they are:

1. Systematic Risk

Systematic risks, or market risks, affect the whole market and cannot be avoided even if investments are diversified. These risks arise mainly because of factors such as:

- Inflation

- Changes in interest rates

- Changes in government policies

- Economic recession

- Global financial crises

The stocks of the banking sector are more vulnerable to systematic risks, and their profitability is directly related to the economic environment and changes in interest rates. This type of risk is usually expressed in terms of Beta in the CAPM model.

2. Unsystematic Risk

Unsystematic risk means risks that are associated with a particular firm. Such risks emanate from internal factors like:

- Management decisions
- Inefficiencies
- Business strategy
- Financial issues related to a particular company

Unsystematic risks differ from systematic risks in that unsystematic risks can be diversified against. Financial theories indicate that only systematic risks result in rewards from the market since investors have an option of diversifying against unsystematic risks. This study has focused on systematic risks only since CAPM is able to measure only systematic risks when estimating the expected return.

2.1.1 Risk–Return Trade-off Theory

The risk-return tradeoff theory asserts that investors demand higher returns in exchange for taking risks. This is the fundamental idea of investment analysis.

For instance:

The government bonds involve low risk and low returns.

Equity shares involve high risks and high returns.

In banking, private sector banks like Axis Bank may involve high risks compared to public sector banks in an unstable economy. Investors would demand high returns for incurring such risks.

Modern Portfolio Theory (Markowitz, 1952)

Modern Portfolio Theory was the theory that introduced diversification and the efficient frontier.

Markowitz proved that it is possible for investors to create an optimal portfolio that will

give the highest return for a certain level of risk.

Unsystematic and systematic risk were also distinguished and formed the basis for asset

pricing models.

Capital Asset Pricing Model (Sharpe, 1964; Lintner, 1965)

According to CAPM:

$$ER_i = R_f + \beta_i (ER_m - R_f)$$

Where,

- Risk-Free Rate = Return on Government Securities
- Beta (β) = Sensitivity of stock return to market movements
- Market Return = Return on market index (NIFTY 50)

Here, Beta measures the sensitivity of stock returns to market movements.

- $\beta = 1 \rightarrow$ Stock moves exactly with the market
- $\beta > 1 \rightarrow$ Stock is more volatile than the market
- $\beta < 1 \rightarrow$ Stock is less volatile than the market

In the case of a bank stock like Axis Bank, Beta calculates the sensitivity of the stock.

Assumptions of CAPM Model

The assumptions of CAPM are:

1. Investors are rational and try to avoid risks.
2. Efficiency of Markets.
3. There are no taxes and transaction costs.
4. Investors have similar expectations.
5. Assets are divisible.

Even though these assumptions are not fully applicable in real-time markets, CAPM is widely accepted due to its simplicity.

Security Market Line (SML)

The Security Market Line is a graphical representation of CAPM. It is a graphical illustration of the relationship between beta, which reflects systematic risk, and expected return.
If a stock is located:

Above the SML → it is undervalued and offers a higher return than required.

Below the SML → it is overvalued and offers a lower return than required.

This concept would help in determining whether Axis Bank shares are fairly valued or not.

Advantages of CAPM

1. User-Friendly to Approach

The primary advantage of the CAPM model is its simplicity. The model requires just a few variables, such as the risk-free rate, beta, and market returns, calculate the returns in security.

2. Focus on Systematic Risk

This type of risk occurs because of changes in the market, such as economic changes, changes in interest rates, and inflation. CAPM helps investors identify the extent to which the stock they are considering is vulnerable to changes in the market, as indicated by its beta value.

3. Useful for Estimating Cost of Equity

The model is largely employed in corporate finance for identifying the cost of equity capital. Corporations use CAPM for investment decisions, projects, and WACC.

4. Widely Accepted in Financial Analysis

The CAPM is well-recognized and accepted in finance, investment, and academic circles. Financial analysts and researchers widely employ this model for assessing the model. The popularity of CAPM made

a widely accepted tool for financial decision-making and analysis.

Limitations of CAPM

1. Unrealistic Assumptions

The CAPM model is built around a series of ideas that just do not add up in terms of how markets really work. For instance, it assumes that investors know everything there is to know, that markets always work efficiently, and that things like taxes and costs of transactions do not even exist. In reality, all of these conditions do not come together at any given time. Hence, it is not surprising that at times the predictions of the model may not be all that solid.

2. Beta is not Fixed

There is another issue with CAPM in that beta is not set in stone. It is constantly in flux depending on how the company is performing, or how the industry is performing, or how the overall economy is performing.

3. Ignores Other Risks

CAPM only examines market risk when it is attempting to predict stock returns. Yet stock prices fluctuate for any number of reasons: company size, level of debt, or even what is going on in the economy. CAPM's failure to consider other risks means it may be missing part of the story when it comes to stock prices.

4. Not Great for Emerging Markets

CAPM is especially difficult to use in emerging markets such as India, where markets may not be smooth and may fluctuate wildly. For instance, unexpected regulations or even an unpredictable economy may interfere with its predictions. CAPM is not very reliable in emerging markets.

CAPM is still an important model even with all its drawbacks.

2.1.4 Efficient Market Hypothesis (Fama, 1970)

The Efficient Market Hypothesis (EMH) asserts that stock prices reflect all the information available. It has three forms:

1. Weak Form
2. Semi-Strong Form
3. Strong Form

The CAP theory assumes semi-strong market efficiency.

Arbitrage Pricing Theory (APT)

Arbitrage Pricing Theory, developed by Stephen Ross, indicates that there are various macro factors, which influence the returns on stocks. Unlike the CAPM, the APT does not rely on any single factor of beta.

The factors would be:
1. Inflation
2. Interest Rates
3. GDP growth rate
4. Exchange rates

Though APT is more flexible, the choice of CAPM in the study has been because of its simplicity and the ability to analyze the performance of a single company.

2.2 Review of Literature

A Studies on CAPM's applicability in emerging economies like India are done using various stock indices like the Nifty 50 and BSE's Sensex. Studies indicate. Although it may not capture all the variations resulting from various economic and stock exchange factors. It is also believed that factors like a change in interest rates, inflation, economic growth, and investor sentiments also impact stock exchange performance. Nevertheless, CAPM is still considered to be a widely used model.

A number of research works done specifically on the banking sector revealed that banking shares are highly volatile in nature in response to various macroeconomic factors, regulatory environment, and interest rate fluctuations. Private sector banks in India, such as Axis Bank, are growing rapidly due to various technological and financial innovations in digital banking and increased involvement in the capital markets. At the same time, this also makes them

vulnerable to market and financial risks. Hence, it is important to understand the relationship between risk and return of banking shares using CAPM theory.

2.3 Analysis of Prior Research Studies

Many research papers have studied over the relevance of CAPM and risk-return trade-off in various markets.

https://www.sciencedirect.com/science/article/abs/pii/S1062976902001254

Sharpe (1964), Lintner (1965) have laid the foundation for CAPM. They have shown a positive correlation between beta and expected returns.

Black, Jensen, and Scholes (1972) have tested CAPM in developed markets and have found some evidence supporting it.

Fama and French (1992) have questioned CAPM. They have stated that beta does not completely explain stock returns. They have suggested considering additional factors.

Merton (1973) has extended CAPM to an intertemporal model.

Studies have also been conducted on emerging markets, specifically India. The studies have shown mixed results. Although CAPM is supported to an extent, there are also macro factors and market inefficiency.

Few studies have specifically looked at Axis Bank Ltd. during and after the pandemic.

2.4 Research Gap

Empirical studies on the link of risk and returns in context of the Indian stock exchange, with the use of models such as the CAPM, APT, and multi-factor models, have been undertaken. Most of the studies are based on the overall market, diversified portfolios, or sectors, and not on an in-depth analysis of individual companies. Although earlier studies on the subject have been useful in providing insights on the applicability of the CAPM model in emerging economies like India, most of the studies were undertaken during stable economic conditions. The economic scenario has undergone tremendous changes over the past few years, particularly after the COVID-19 pandemic, which has created unprecedented changes in the global supply chain, consumer behavior, and market volatility.

For instance, in India, the banking industry was facing credit demand fluctuations, asset quality pressures, loan restructuring, and rapid digitalization of banks. These may have impacted the risk and return of banking stocks.

In addition, in recent times, there have been regular announcements of changes in interest rate policies by the Reserve Bank of India (RBI), rising inflationary trends, geopolitical events, and rising global financial integration.

These macro-economic factors have a direct impact on the performance of the banking industry since banks tend to be highly sensitive to interest rate and credit policy changes. There is rapid digitalization in the banking industry in India through partnerships with fintech companies and rapid technological innovations in private sector banks.

Despite such major developments in the financial markets of India and the world over, limited research in recent times has focused specifically on reassessing the relationship between risk and return of major private sector banks in India using updated data. There is particularly limited focused empirical work undertaken with respect to Axis Bank Ltd., which is one of the key constituents of the Nifty 50 index and one of the major private sector banks in India. Most of the existing literature either focuses on analyzing multiple banks under one study for the broader category of banks or relies on data that may not be relevant in today's context.

Thus, there is definitely a research gap in reassessing whether or not the traditional CAPM model is applicable in determining the expected rate of return for Axis Bank Ltd. in today's context. This study aims to fill in that research gap by reassessing the systematic risk (Beta) and expected rate of return of Axis Bank Ltd. using updated historical data.

2.4 Conceptual Understanding

The conceptual framework of this study is based on the CAPM model.

Independent Variable:

Beta

Dependent Variable:

Expected Return

Control Variables:

Risk-Free Rate and Market Return

The research aims to find whether beta of Axis Bank is significant in determining its expected return as per the CAPM model.

If the original Return is:

• Greater than E(R) → Undervalued Stock

• Less than E(R) → Overvalued Stock

CHAPTER 3: RESEARCH METHODOLOGY

3.1.1 Research Design

For this purpose, the current research is based on a quantitative, descriptive, and analytical research design to examine the risk and return of Axis Bank Ltd. The quantitative approach is appropriate because it is based on actual data, statistical calculations, and financial modeling. The research does not require opinions and is based on actual data available in the stock market.

This research is based on actual data available in financial markets. The data includes actual systematic risk, calculation of average returns, and actual returns compared with expected returns using the CAPM model. The actual data available in financial markets add more credibility to this research.

The study will also be descriptive, with the analysis of the historical price and returns of the Axis Bank stock over a given period. This will be done by analyzing the monthly stock prices of the company from January 2021 to December 2024, giving an accurate idea of how the stock has performed compared to the overall market index (NIFTY 50).

The research will be analytical, moving beyond the simple description of the data and the use of statistical methods to interpret the data on the financial relationships. This will be done by calculating the beta coefficient using regression analysis with the Excel SLOPE function, followed by the CAPM equation to evaluate if the stock has generated enough returns compared to the risks involved.

The time period of four years, i.e., 2021 to 2024, has been considered to match the changing trends of the market, such as the recovery of the market after the pandemic, the change in interest rates, and the changing market conditions. Monthly data has been considered over daily data to avoid any short-term changes and to ensure the reliability of the analysis.

The purpose of the research design is to assess the effectiveness of the CAPM theory in explaining the risk-return tradeoff of Axis Bank during the considered period. The analysis of the historical data and the financial theory will enable the reader to gain significant insights into the performance of the stock.

3.1.2 Data Sources and Justification

The research only used secondary data collected from reliable financial sources.

1. The monthly closing price of Axis Bank Ltd was collected from financial sources such as Yahoo Finance, NSE, and Investing.com.

2. The monthly closing values of the NIFTY 50 Index were used as proxies for market returns.

3. The risk-free rate was assumed to be 6% annually, which is the average 91-day Treasury Bill Rate from the Reserve Bank of India (RBI), approximately 0.5% on a monthly basis.

Justification for Using Secondary Data:

1. Stock market data is historical and publicly available.

2. CAPM analysis requires past market returns and price data.

3. Secondary data helps to ensure reliability and authenticity.

4. It is affordable and saves time for academic research.

5. Financial databases provide precise and standardized data for statistical analysis.

Therefore, the use of secondary data is appropriate and justified for conducting risk-return analysis.

The approach followed for this study is to evaluate the risk-return profile of Axis Bank Ltd using the Capital Asset Pricing Model. This study is based on secondary data collected from credible financial data sources such as Yahoo Finance and National Stock Exchange (NSE). Monthly closing prices of Axis Bank Ltd shares and Nifty 50 index shares were obtained from January 2021 to December 2024, for a total of four years.

3.1.3 Analytical Tools Used

The study employs the following analytical tools:

1. Calculation of Monthly Returns

The monthly returns for Axis Bank and NIFTY 50 were computed using the formula:

$$Return = \frac{(P_t - P_{t-1})}{P_{t-1}}$$

Where:

Pt = Price of current month

Pt-1 = Price of previous month

This formula calculates percentage change in stock price over two consecutive months.

2. Beta Estimation

Beta is an important factor used in financial analysis. For purpose of this study, the value of beta for Axis Bank Ltd has been calculated with the help of the SLOPE function in Microsoft Excel. This function helps us estimate the relationship between two sets of variables.

Beta = SLOPE (Axis Bank Returns, Market Returns)

In the formula, the returns of Axis Bank will be the dependent variable, and the Market Returns (Nifty 50) will be the independent variable. This method of statistics is generally used to determine the beta value because it indicates the linear relationship between the performance of stocks and the overall performance of the market.

The determined value of beta indicates the extent to which the returns on the stocks of Axis Bank are affected by the changes in the market index. If the value of beta equals 1, then the stocks are moving in the same direction as the market. Also, if the value of beta is more than 1, then the stocks are more volatile than the market. This means the stocks will grow more than

the market and vice versa. Thus, if the value of beta is less than 1, then the stocks are less volatile than the market, which means the stocks are more stable than the market index.

In this particular study, the beta value calculated for Axis Bank reflects the systematic risk involved in investing in the shares of this bank compared to the general stock market, reflected in the value of the Nifty 50 index.

By utilizing Excel SLOPE function, there is a reliable method to statistically estimate the beta value, and this assists in determining whether the returns of Axis Bank are in line with those calculated in the CAPM theory.

3. Application of CAPM

The Capital Asset Pricing Model has been used to calculate the expected returns as follows:
Expected Return=Rf+β(Rm−Rf)
Where:

- R_f = Risk-free rate (0.005 monthly)
- R_m = Average market return
- β = Beta of Axis Bank

The monthly returns of Axis Bank were calculated on the basis of the percentage change in the adjusted closing prices. Similarly, the returns on the market were calculated on the basis of the percentage change in the values of the Nifty 50 index. These returns were then used to calculate the systematic risks of Axis Bank with the help of coefficient.

In the present study, the beta value has been calculated with the help of the SLOPE function in Microsoft Excel. This function indicates the relationship between the returns on stocks and the returns on the market. Once the value of the beta coefficient has been obtained, the expected returns on Axis Bank stocks can be obtained with the help of the Capital Asset Pricing Model. It has been assumed that the rate of interest on the risk-free assets, i.e., the 91-day Treasury Bill issued by the Reserve Bank of India, would be 6% annually.

4. Software Used

Microsoft Excel was used for:

- Data organization
- Monthly return calculations
- Beta estimation
- CAPM computation
- Preparation of tables
- Graphical representation (line charts and scatter plots)

3.1.4 Limitations of the Study

The study has several limitations:

1. The analysis is only for a four-year period from 2021 to 2024.

2. The analysis is only for Axis Bank Ltd and does not take into account other banking stocks.

3. CAPM only accounts for systematic risk and does not account for unsystematic risk.

4. The risk-free rate is assumed to be constant over the analysis period.

5. Market shocks and other factors can influence the return.

Overall, despite the limitations, the study has provided valuable insights on the risk-return behavior of Axis Bank.

CHAPTER 4: DATA ANALYSIS AND INTERPRETATION

3.2 DATA ANALYSIS AND INTERPRETATION

3.2.1 Summary of Empirical Findings

Particulars	Value
Beta (β)	1.1998
Average Market Return (Rm)	0.0124
Risk-Free Rate (Rf)	0.005
Actual Average Return (Axis)	1.24%
CAPM Expected Return	1.39%

Table -01

3.2.2 Analysis of Beta

The beta value of 1.1998 indicates that the bank is more volatile compared to the market. When the beta value is more than one, the stock is more volatile compared to the market. This means that:

1. When the market goes up, the stock of Axis Bank goes up more than the market.
2. When the market goes down, the stock of Axis Bank goes down more than the market.

This indicates that the bank has more systematic risk compared to the NIFTY 50 index.

3.2.3 CAPM Expected Return Analysis

1. By using the CAPM model, we have calculated that expected rate of return for Axis Bank is 1.39%.
2. The actual rate of return earned by Axis Bank over the period of study was 1.24%.
3. As the actual rate of return is less compared to the expected rate of return.
4. Axis Bank slightly underperformed compared to its required rate of return in terms of risk.

5. This difference of 0.15% in actual and expected rate of return shows that investors were not compensated for taking extra risk.

3.2.4 Comparative Interpretation

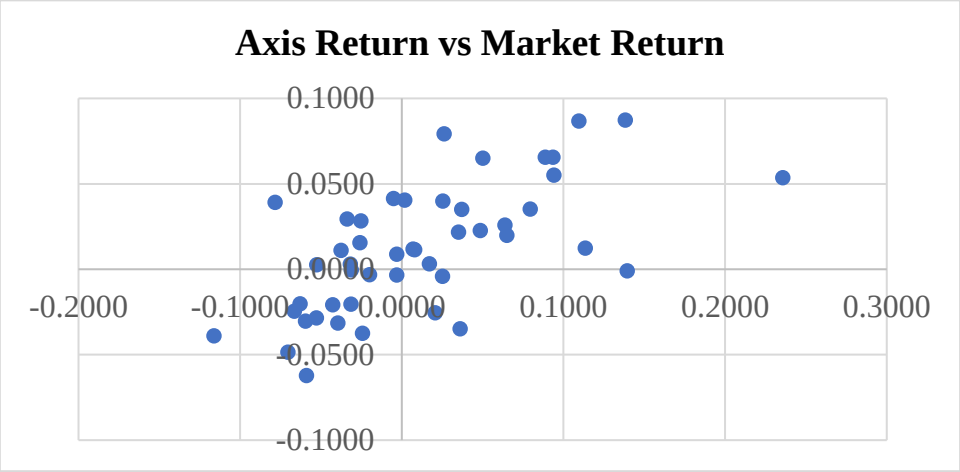
Although the average market return and the return of Axis Bank were the same, at 1.24%, the beta value indicates that there was more volatility in Axis Bank. This means, according to CAPM theory, investors expected a higher return than the average market return.

The reason the company has failed to generate returns above the average market return is:

- Market inefficiency
- Impact of macroeconomic factors
- Impact of economic changes following the pandemic
- Challenges facing the banking sector

3.2.5 Graphical Representation

The following charts were prepared to support the findings:



Scatter Plot

Figure-01

Interpretation

1. The scatter plot shows a positive relationship between Axis Bank returns and market returns.
2. The upward sloping regression line confirms that Axis Bank's returns move in the same direction as the market.
3. The estimated beta value of 1.1998 indicates that Axis Bank is more sensitive.
4. Beta is greater than one which is market goes up, Axis Bank's returns tend to increase more. When the market goes down, the stock may drop more sharply.
5. The spread of data points around the regression line suggests there is company-specific risk alongside market risk.
6. Overall, the scatter plot visually supports the calculated beta value and confirms that Axis Bank has higher market risk compared to the market.

Line chart

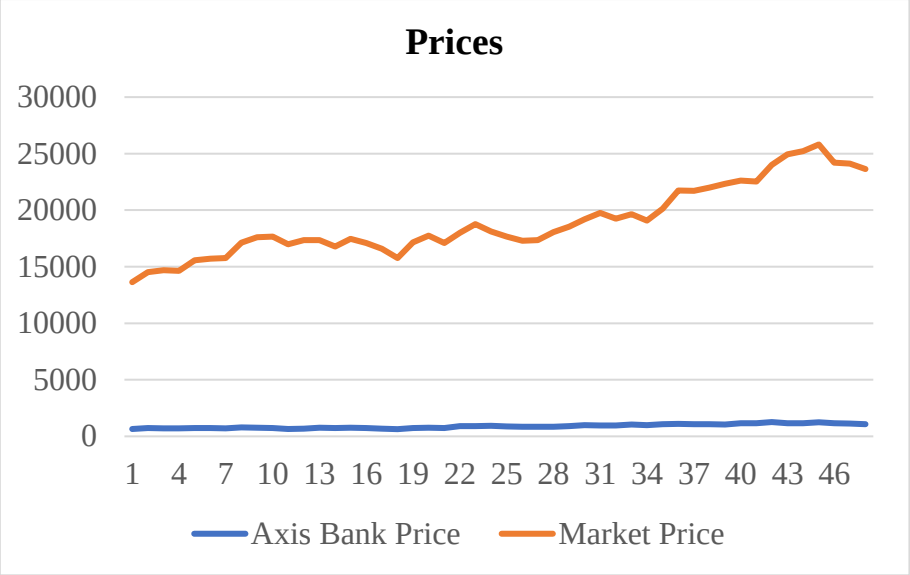


Figure -02

Interpretation:

1. The line chart displays the changes in Axis Bank stock prices and the NIFTY 50 index from 2021 to 2024.
2. 2. The NIFTY 50 index has shown significant growth over the entire period.
3. 3. Axis Bank has also shown price growth over the entire period, albeit at a lower rate.
4. 4. Axis Bank's price has shown fluctuations, indicating its response to market movements.
5. 5. The price movement has been consistent with the beta value of 1.1998.
6. 6. The visual analysis suggests that although Axis Bank has shown growth along with the market, it has not shown higher growth than the market over the entire period.

3.2.6 Key Findings

1. Axis Bank also reflects a high level of systematic risk, as indicated by $\beta = 1.1998$.

2. CAPM also calculated an expected return of 1.39%.

3. The actual return is 1.24%.

4. Axis Bank underperformed slightly compared to the expectations of CAPM.

5. Based on this result, it is indicated that higher risk did not proportionally increase returns in 2021 to 2024.

CHAPTER 5: FINDINGS AND CONCLUSION

4.1 Findings

The results of the study indicate that Axis Bank's beta is higher than one, which means that it is a riskier investment than others in the market. This indicates that the stock is more sensitive to market risks. Financial theories, particularly the CAPM, suggest that a high-risk investment is likely to generate high returns in order to compensate for the risks involved.

This research has significant implications with regard to the systematic risk and expected return relationship for Axis Bank Ltd. The beta value is higher than one, which shows that Axis Bank stock reacts more to market movements, and hence the returns on Axis Bank stock vary more than the returns on the market as a whole when the market moves up and down. The findings of this research are in line with financial theory, which states that investors require a premium

to undertake more risk, and hence stocks with more systematic risk provide investors with higher expected returns.

This has been shown through the fact that there is an agreement between average return of Axis Bank and predicted return using the CAPM model. This shows that it is a good reflection of the actual performance of the stock over this period and therefore gives strong evidence of the continued use of this model in assessing risk and return within this Indian banking sector. In addition, it has identified other factors which can impact the stock market, such as the economy, issues related to regulations, and issues related to technology and investors.

For management, this study has shown that it can be of great use to investors and financial analysts in assessing the risk profile of Axis Bank's shares. It can also be of great use to Portfolio Managers in deciding on possible investment alternatives and in building up a portfolio which can offset risk and return. Thus, this study has shown it can be of great use in enhancing the overall understanding of investment decision-making as it relates to the Indian stock market.

However, from the results, it is evident that even though Axis Bank faces a higher level of market risk compared to other firms, its average return over this period is slightly lower than what CAPM theory predicts. It is evident that the expected relationship between risk and return as provided by CAPM did not totally materialize.

These results are in line with existing financial literature that points out that CAPM, although widely accepted, may not completely capture stock returns in emerging markets such as India.

From 2021 to 2024, there were various external factors such as recovery from the pandemic, changing interest rates, inflation, and banking regulations. All these factors could have impacted Axis Bank's return pattern, thus creating a mismatch in expected and actual returns.

Although there is a positive correlation between market returns and Axis Bank's returns, validating the CAPM theory to a certain extent, there is a minor difference, indicating that there are other factors that are also influencing stock returns.

4.2 Conclusion

The study considered the Risk and Return relationship of Axis Bank Ltd listed on the National Stock Exchange, and the CAPM model has been used for the period ranging from 2021 to 2024.

The key findings of the study are:

• Axis Bank has more systematic risks compared to the market.

• The stock has more market response.

• The CAPM model has estimated the expected returns, which are higher because of the higher beta value.

• The stock has slightly underperformed compared to the CAPM model.

The analysis indicates that the impact of systematic risks on stock returns is significant, which is in line with the CAPM theory. However, the slight underperformance indicates that the CAPM model might not be appropriate for the banking industry. So, it’s a useful kit, and the real-world market conditions and the dynamics need to be considered.

4.3 Managerial and Policy Implications

The implications of the study's findings are relevant for investors, financial managers, and policymakers:

Investors:

1. Investors must consider beta while making investments, particularly in banking stocks.

2. Beta stocks like Axis Bank may offer higher potential returns but may also expose investors to higher market risks.

3. Portfolio managers must diversify investments to manage systematic and unsystematic risks.

4. The performance of banking stocks is not only dependent on market risks but is also subject to various regulatory policies and interest rate fluctuations.

5. For policymakers:

Regulatory bodies must ensure financial stability and regulations to avoid excessive market volatilities.

Thus, it is clear that the importance of risk and return analysis is of considerable use in making investment decisions.

4.4 Future Research Directions

Though the study provides good insights, further studies could be done to expand the analysis in the following ways:

1. More years could be included in the data to obtain a broader view.

2. More stocks of the banking sector could be considered instead of just one company.

3. Multi-factor models such as APT or FF Models could be used.

4. Macro variables like inflation, GDP growth rate, and interest rates could be included.

5. Daily data could be used to analyze short-term volatility patterns.

Further studies taking these factors into account could lead to more insights into the stock returns in the Indian capital market.

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